

Chapter 5

Paninian Grammar

1

5.1 Introduction

How is it that natural language is used by speakers to convey information to the hearers? How is it that on hearing an utterance, the hearers are able to get the intended information? These are the questions that have intrigued Paninians. The goal of the Paninian approach is to construct a theory of human natural language communication that answers these questions. Grammar, a part of such a theory of communication, is a system of rules that establishes a relation between what the speaker decides to say and his utterance, and similarly, what the hearer hears and the meaning he extracts.²

The main problem the Paninian approach addresses is how to extract karaka relations (which are syntactico-semantic relations) from a sentence. As it is inspired by an inflectionally rich language, it emphasizes the roles of case endings and markers such as post-positions (or prepositions). Positions or word order is brought into consideration only when necessary.

A majority of human languages including Indian and other languages have relatively free word order. In free word order languages, order of words contains only secondary information such as emphasis etc. Primary information relating to ‘gross’ meaning (e.g., one that includes semantic

¹K.V. Ramakrishnamacharyulu of Rashtriya Sanskrit Vidyapeetha, Tirupati is the co-author of this chapter.

²There is a difference between the goals of Paninian approach on the one hand and (Chomskyan) generative enterprise on the other. The latter is interested in identifying the innate universal grammar in the mind of every human child by which he or she so effortlessly and without explicit tutoring acquires language that he or she is exposed to.

relationships) is contained elsewhere. In contrast to the languages, most existing computational grammars are based on context free grammars which are basically positional grammars. These are designed for languages in which position of a constituent in a sentence contains key information. It is important to develop a suitable computational grammar formalism for free word order languages for two reasons:

1. A suitably designed formalism will be more efficient because it will be able to make use of primary sources of information directly.
2. Such a formalism is also likely to be linguistically more elegant and satisfying. Since it will be able to relate to primary sources of information, the grammar is likely to be more economical and easier to write.

In this chapter, we describe such a formalism, called the Paninian framework, that has been successfully applied to Indian languages.

5.2 The Semantic Model

In this section, we describe the elements of the semantic model in the Paninian framework.

Every verbal root (dhaatu) denotes an action consisting of:

1. an activity (a vyaapaara), and
2. a result (a phala)

Result is the state which when reached, the action is complete. *Activity* consists of actions carried out by different participants or *karakas*³ involved in the action.

An action is usually a complex which can be broken into sub-actions. For example, in the opening of a lock, a person inserts a key in the lock and turns it, the key on its part presses the levers and moves them, the latch in turn moves and the lock opens (Figure 5.1). Each of the sub-actions (e.g., inserting and turning a key, the key pressing and moving the levers, the latch moving and the lock opening) has its own semantic relations with associated objects.

Another important concept is that of vivaksha. *Vivaksha* refers to the speaker's viewpoint or attitude towards the activity. A sentence is not only a statement of an objective activity but also contains information regarding the speaker's viewpoint. (Vivaksha should not be confused with the

³Karaka pronounced as 'kaarak' is defined later.

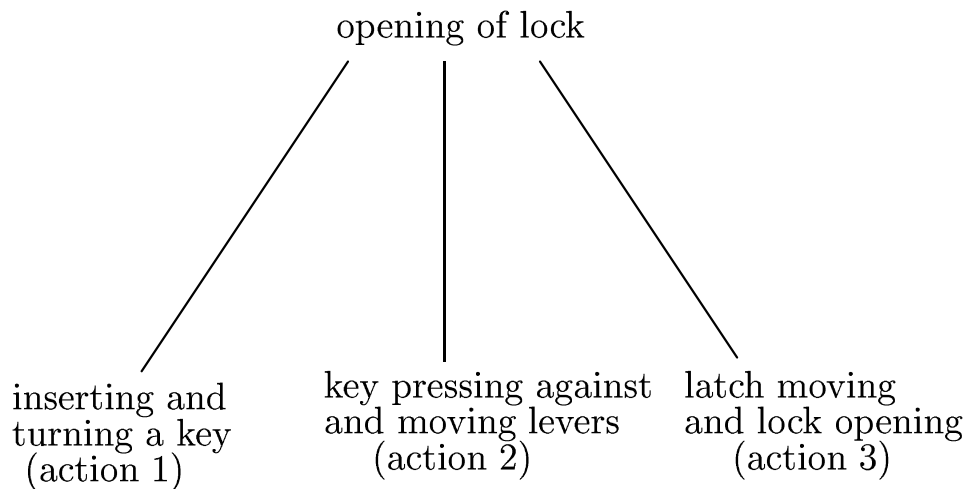


Figure 5.1: Structure of an action

speaker's intentions which come under pragmatics.) Usually, *vivaksha* affects the choice of verb form which, in turn, affects the choice of participants and their relation with or role in the action.

Out of all the participants in an action, there is one who is 'swatantra' or most independent, and is called *karta*⁴ *karaka*. Thus, in the following sentences:

- A.1 The boy opened the lock.
 A.2 The key opened the lock.
 A.3 The lock opened.

the *karta* is, respectively, the boy, the key, and the lock.

The Paninian theory chooses to talk in terms of *karta* instead of agent. It is not just a change in nomenclature, but as the above example shows, *karta* and agent are different in general.

According to Kaundbhata, who further elaborates on the concept of *swatantra*, every verb in a sentence refers to a complex of activity. Frequently, however, the same verb can be used to refer to not only the main activity but also to sub-parts of the complex.⁵ *Karta* of a verb in a sentence is one which is the 'aashraya' or locus of the activity. Thus, in A.1 the activity referred to is the act of opening of the lock by the boy by inserting a key and turning it, etc. (action 1 in Figure 5.1). Thus, the boy is the

⁴Karta is pronounced 'kartaa'.

⁵Sometimes the verb can be ambiguous, in which case it may refer to one of several actions. When used in a sentence, it may refer to any one of its possible actions.

karta of this activity. In A.2, the activity referred to is the pressing and moving of the levers by the key causing the lock to open, etc. (action 2 in Figure 5.1). The ‘key’ is the karta of this activity. In A.3, the activity is that of motion of the latch and opening of the lock (action 3 in Figure 5.1). Here, the karta is the lock. In A.1 the speaker decides to give importance to the role of the boy, whereas in A.2 the speaker wishes to emphasize the role of the key. (“It is *this* key which opened the lock!”) In A.3 the speaker emphasizes the fact that the lock has been opened; which key opened it and who opened is unimportant as far as that utterance is concerned. As further evidence that the same verb ‘open’ refers to distinct actions (however related they might be), note that in a language like Hindi there is a different word for ‘open’ in A.3 as compared to A.1 and A.2. See B.1, B.2 and B.3 below for Hindi sentences.

B.1 mohana ne taalaa kholaa.
 Mohan -ne lock opened
 (Mohan opened the lock.)

B.2 isa caabii ne taalaa kholaa.
 This key -ne lock opened
 (This key opened the lock)

B.3 taalaa khula gayaa.
 lock opened
 (The lock opened.)

The notion of karaka relations is central to the Paninian model. The karaka relations are syntactico-semantic (or semantico-syntactic) relations between the verbals and other related constituents in a sentence. They capture a certain level of semantics. But this is the level of semantics that is important syntactically and is reflected in the surface form of the sentence(s). Figure 5.2 shows the relationship.

In Figure 5.2, the surface level is the uttered or written sentence. The vibhakti level is the level at which there are local word groups based on case endings, preposition or postposition markers.⁶ Vibhakti for nouns has already been defined earlier. A noun group is a unit containing a noun (or a pronoun, proper name, etc.), its vibhakti and possibly some adjectives.

Vibhakti for verbs can be defined similar to that for the nouns. A head verb may be followed by auxiliary verbs (which may remain as separate words or may combine with the head verb). Such information is collectively called vibhakti for the verb. The vibhakti for a verb gives information about tense, aspect and modality (TAM), and is, therefore, also called the TAM label. TAM labels are purely syntactic determined from the verb form and the auxiliaries.

⁶For positional languages such as English, it would also include position or word order information.

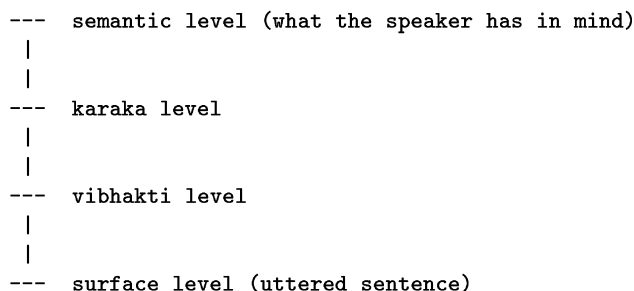


Figure 5.2: Levels in the Paninian model

The vibhakti level abstracts away from many minor (including orthographic and idiosyncratic) differences among languages. The topmost level relates to what the speaker has in his mind. This may be considered to be the ultimate meaning level. Between this level and vibhakti level is the karaka level. It includes karaka relations and a few additional relations such as tadarthya (or purpose). One can imagine several levels between the karaka and the ultimate level, each containing more semantic information. Thus, karaka level is one in a series of levels, but one which has relationship to semantics on the one hand and syntax on the other.

There are only around six different types of karaka relations. It is clearly not possible for them to capture the innumerable types of semantic relations among all possible actions or states (written as verbs) and all possible objects (written as nominals) in the world. What the karaka relations do specify with respect to a verb, however, are six or so relations of nominals (or verbals) that participate in the action specified by that particular verb. These are sufficient for providing a mapping from karaka relations to semantic relations. Thus, the karakas provide the maximum necessary information relative to a verb. As an example, the karta karaka may get mapped to agent for one verb, and experiencer for another, etc.

We have already seen the definition of karta karaka. Here, we informally discuss the remaining ones. We saw earlier that karta, the swatantra or the most independent participant in an action, is the ashraya (or the *locus*) of activity. In other words, the activity resides in or springs forth from the karta. Ashraya of the result is called karma, if it can be different from karta. A verb in which ashraya of activity and result can be different, is called *sakarmaka* (loosely, *transitive*). For example, in the following sentence:

raama ne raavana ko tiira se maaraa.

Ram ergative Ravan accus. arrow instr. killed
 (Ram killed Ravan with an arrow.)

Ram is the karta, and Ravan is karma.

Similarly karana karaka is the instrument. With the vyapara (or activity) of the karana, phala (or result) is immediately achieved. Thus, in the above example, with the vyapara of the arrow (i.e., motion of the arrow and striking Ravan) the result of the verb (kill) is immediately achieved.

Adhikarana karaka is the locus of karta or karma. It is what supports, in space or time, the karta or the karma.

Apadan ⁷ karaka is that participant in an action involving separation which remains stationary (or is the reference) when the separation takes place. Note that vivaksha or speaker viewpoint towards a separation event is what is important in determining what is stationary or the reference point. For example, in the following sentence:

peDa se baaga meiM patte gire.
 tree -se garden -meiM leaves fell
 (Leaves fell from the tree in the garden.)

the tree is the apadana, and garden is adhikarana.

Sampradana ⁸ is the beneficiary. For example, in the following sentence:

laDakaa laDakii ko phoola detaa hei.
 boy girl acc. flower gives
 (The boy gives a flower to the girl.)

the ‘girl’ is the sampradana.

There is another relation called tadarthya (roughly translated as purpose), which connects two verbs by the ‘purpose’ relation. For example:

aadami ne naava banane ke liye lakaDii kaaTii.
 man -ne boat to-make -ke -liye wood cut
 (The man cut wood to make a boat.)

Even though taadaarthya is not a karaka relation, it is included here because of the important role it plays.

It should be re-emphasized that other than karta, we have had no detailed discussion on any other karaka. For more details on other karakas see work by Kaundabhatta (Shastri, 1973).

5.3 Free Word Order and Vibhakti

Indian languages have a relatively free word order. Many of the constituents of a simple sentence can occur in any order without affecting the gross

⁷Pronounced ‘apaadaana’

⁸Pronounced ‘sampradaana’

meaning of the sentence; what is affected is perhaps the emphasis, etc. For instance, noun groups in a sentence can come in any order without affecting the karaka relationship (or theta relationships) between the verb and the noun groups. Since position or order of occurrence of a noun group does not contain information about the karaka or theta roles in a simple sentence, a question can be asked regarding what carries this information. The answer seems to be that post-position markers after nouns (in North Indian languages) or surface case endings of nouns (in South Indian languages) or a mixture of the two at times, play a key role in specifying semantic relationships. We will collectively refer to the post position markers and surface case endings of nouns as vibhakti of nouns. Consider the following sentences of Hindi (Bharati et al. (1991a)).

C.1 raama mohana ko piiTataa hei.
 Ram Mohan -ko beat is
 (Ram beats Mohan.)

C.2 mohana ko raama piiTataa hei.
 Mohan -ko Ram beat is
 (Ram beats Mohan.)

C.3 mohana raama ko piiTataa hei.
 Mohan Ram -ko beat is
 (Mohan beats Ram.)

C.4 raama ko mohana piiTataa hei.
 Ram -ko Mohan beat is
 (Mohan beats Ram.)

In C.1 and C.2, Mohan has the same vibhakti (i.e., *parsarg* or postposition 'ko') and semantic relation with beating. Even though the positions of 'raama' and 'mohana ko' are interchanged in C.2, it does not alter the respective semantic relations of Ram and Mohan with the verb. C.3 and C.4 show that semantic relation of Ram is interchanged with that of Mohan by interchanging their vibhakti. So the vibhaktis are crucial in determining the semantic roles. Relative position of the nominal seems to be not very important for determining semantic relations.

However, things are not always straightforward and the following need to be accounted for:

1. A different vibhakti can be used for the same semantic relation with a given verb in a different sentence. For example, in the Hindi sentences D.1 to D.4, although Ram has the same semantic relation with eat (namely, the agent of eat), a different vibhakti is used each time (ϕ , ne, ko, se which represent nominative, ergative, accusative, ablative, respectively).

- D.1 raama phala ko khaataa hei.
 Ram fruit -ko eats is
 (Ram eats the fruit.)
- D.2 raama ne phala khaayaa.
 Ram -ne fruit ate
 (Ram ate the fruit.)
- D.3 raama ko phala khaanaa paDaa.
 Ram -ko fruit eat had-to
 (Ram had to eat the fruit.)
- D.4 raama se phala nahii khaayaa gayaa.
 Ram -se fruit not eat could
 (Ram could not eat the fruit.)

2. Same vibhakti can be used with the same verb for two different semantic relations. In E.1 and E.2, the same vibhakti (that is, 'ne' parsarg) is used for Mohan and the key, whereas their semantic relation to 'open' are quite different. Similarly for 'se' in E.3 and E.5, (E.1, E.2 and E.4 are same as B.1, B.2, B.3, respectively).

- E.1 mohana ne taalaa kholaa.
 Mohan -ne lock opened
 (Mohan opened the lock.)
- E.2 isa caabii ne taalaa kholaa.
 this key -ne lock opened
 (This key opened the lock)
- E.3 caabii se taalaa khula gayaa.
 key by lock opened
 (The lock opened 'unexpectedly' by the key.)
- E.4 taalaa khula gayaa.
 lock opened
 (The lock opened.)
- E.5 bacce se taalaa khula gayaa.
 child by lock opened
 (The lock was opened unintentionally by the child)
 OR (The child was able to open the lock.)

5.4 Paninian Theory

Paninian grammar is particularly suited to free word order languages. It makes use of vibhakti information for mapping to semantic relations, and uses position information only secondarily. As the Indian languages have (relatively) free word order and vibhakti, they are eminently suited to be described by Paninian grammar. The Paninian framework was originally designed more than two millennia ago for writing a grammar of Sanskrit; it has been adapted by us to deal with modern Indian languages.

5.4.1 Karaka Relations

Example sentences in Hindi from the previous section (in C, D and E) indicate that there is no straightforward mapping from vibhakti to semantic relation between noun groups and verbs. The key to arriving at an answer is to identify appropriate relations, i.e., karaka relations.

The most important insight regarding the ggkaraka-vibhakti mapping is that it depends on the verb and its tense aspect modality (TAM) label. The mapping is represented by two structures: default karaka chart and karaka chart transformation. The default karaka chart for a verb or a class of verbs gives the mapping for the TAM label known as basic. It specifies the vibhakti permitted for the applicable karaka relations⁹ for the nouns etc. when the verb has the basic TAM label. The basic TAM label chosen for Hindi is ‘taa.hei’ and roughly corresponds to present indefinite tense. Any other TAM label could have been chosen as basic without any problem. The TAM labels are purely syntactic in nature and can be determined by looking at the verb form and the associated auxiliary verbs, etc. For other TAM labels, there are karaka chart transformation rules. Thus, for a given verb with some TAM label in a sentence, appropriate karaka-vibhakti mapping can be obtained using its default karaka chart and the transformation rule depending on its TAM label.

The default karaka chart for three of the karakas is given in Figure 5.3. This explains the vibhaktis in sentences C.1 to C.4. As Ram is the agent in C.1 and C.2, and Mohan in C.3 and C.4, and the agent is the most independent for the action (beat), it is expressed by means of the karta karaka; and the remaining nominal by karma karaka. For the karta karaka, ϕ vibhakti is used with taa.hei TAM label in C.1 to C.4 as explained in Figure 5.3.

Figure 5.4 gives some transformation rules for the default mapping for Hindi. It explains the vibhakti in sentences D.1 to D.4 (assuming that Ram is the karta and phala (fruit) is the karma). As explained by Figure 5.4,

⁹What karaka relations are permissible for a verb, obviously depends on the particular verb. Not all verbs will take all possible karaka relations.

<i>Karaka</i>	<i>Vibhakti</i>	<i>Presence</i>
Karta	ϕ	mandatory
Karma	ko or ϕ	mandatory
Karana	se or dvaaraa	optional

Figure 5.3: Default karaka chart

karta takes ‘ne’ vibhakti in D.2 because of TAM label ‘yaa’ (in the main verb group khaayaa), ‘ko’ in D.3 because of TAM label naa_paDaa (in khaanaa paDaa), and ‘se’ in D.4 because of TAM label yaa_gayaa (in khaayaa gayaa). (See Appendix for how Paninian Grammar handles similar phenomena in Sanskrit.)

<i>TAM label</i>	<i>Transformed vibhakti for karta</i>
yaa	ne
naa_paDaa	ko
yaa_gayaa	se or dvaaraa (and karta is optional)

Figure 5.4: Transformation rules

The default mapping and transformation rules also explain E.1 to E.5. Note that by our definition of swatantra or independent, Mohan is the karta in E.1, caabii (or key) in E.2, and taalaa (or lock) in E.3, E.4 and E.5. After this the vibhaktis of karta in E.1 to E.5 are as explained in Figure 5.4.¹⁰

In general, the transformations affect not only the vibhakti of karta but also that of other karakas. They also ‘delete’ karaka roles at times, that is, the ‘deleted’ karaka roles do not occur in the sentence.

It is important to re-emphasize that the transformation depends on TAM label which is purely syntactic, and not on tense, aspect and modality which are semantic. The TAM label can be determined for Hindi and other Indian languages by syntactic forms of the verb and its auxiliaries without the need to refer to any semantic aspects. The specification for obtaining TAM labels is given by a finite state machine as described in Chap. 4.

The Paninian theory outlined above (i.e., karaka charts and karaka chart transformations) can be used to generate (or analyze) the sentences given above as we have seen. However, there are additional constraints that would disallow the following sentences to be generated, for example¹¹:

¹⁰The choice of nominals to be used in the sentence is made by the speaker using vivaksha while mapping from semantic level to the karaka level.

¹¹A ‘*’ before a sentence indicates that it is not a good sentence.

F.1 *ladake ne raama ne laDakii ko kitaaba dii.
 boy -ne Ram -ne girl -ko book gave
 (*The Ram the boy gave a book to the girl.)

F.2 *laDake ne laDakii ko kitaaba phoola dii.
 boy -ne girl -ko book flower gave
 (*The boy gave a book a flower to the girl.)

The constraints are given below:

1. Each mandatory karaka in the karaka chart for each verb group, is expressed *exactly once*. (In other words, a given mandatory karaka generates only one noun group with the specified vibhakti in its karaka chart unlike in F.1.)
2. Each optional karaka in the karaka chart for each verb group, is expressed *at most once*.
3. Each source word group satisfies some karaka relation with some verb (or some other relation). In other words, there should no unconnected source word group in a sentence, otherwise, the sentence becomes bad as in F.2.

Karaka charts are based on the idea of aakaankshaa and yogyataa mentioned earlier in Chap. 2. A karaka chart for a verb expresses its aakaankshaas or demands, and specifies the vibhaktis that must be used (i.e., yogyataa) with word groups that satisfy the demands.

The same ideas can be used to handle noun-adjectives, noun-noun relations, verb-verb relations etc.

To show that the above, though neat, is not just an ad hoc mechanism that explains the isolated phenomena of semantic roles mapping to vibhaktis, we discuss two other phenomena: active-passive and control.

5.5 Active Passive

No separate theory is needed to explain active-passives. Active and passive turn out to be special cases of certain TAM labels, namely, those used to mark active and passive. Again consider the following example in Hindi.

F.3 raama mohana ko piiTataa hei. (active)
 Ram Mohan -ko beats pres.
 (Ram beats Mohan.)

F.4 raama dvaaraa mohana ko piiTaa gayaa. (passive)
 Ram by Mohan -ko beaten was
 (Mohan was beaten by Ram.)

Verb in F.4 has TAM label as yaa-gayaa. Consequently, the vibhakti ‘dvaaraa’ for karta (Ram) follows from the transformation already given earlier in Figure 5.4.

5.6 Control

A major support for the theory comes from complex sentences, that is, sentences containing more than one verb group. We first introduce the problem and then describe how the theory provides an answer. Consider the following sentences in Hindi:

G.1 raama phala khaakara mohana ko bulaataa hei .
 Ram fruit having-eaten Mohan -ko calls is
 (Having eaten fruit, Ram calls Mohan.)

G.2 raama ne phala kaaTakara khaayaa .
 Ram ne fruit having-cut ate
 (Ram ate having cut the fruit.)

G.3 phala kaaTane ke liye usane caakuu liyaa .
 fruit to-cut -ke-liye he-ne knife took
 (To cut fruit, he took a knife.)

In G.1, Ram is the karta of both the verbs: khaa (eat) and bulaa (call). However, it occurs only once. The problem is to identify which verb will control its vibhakti. In G.2, karta Ram and the karma phala (fruit) both are shared by the two verbs kaaTa (cut) and khaa (eat). In G.3, the karta ‘usa’ (he) is shared between the two verbs, and ‘caakuu’ (knife) the karma karaka of ‘lii’ (take) is the karana (instrumental) karaka of ‘kaaTa’ (cut).

The observation that the matrix or main verb rather than the intermediate verb controls the vibhakti of the shared nominal is true in the above sentences. The theory we will outline to elaborate on this theme will have two parts. The first part gives the karaka to vibhakti mapping as usual, the second part identifies shared karakas.

5.6.1 Karaka to Vibhakti Mapping

The intermediate verbs have their TAM labels just like other verbs. For example, kara is the TAM label of khaa (eat) in G.1 and G.2, and naa is the TAM label of kaaTa (cut) in G.3. As usual, these TAM labels have transformation rules that operate and modify the default karaka chart. In particular, the suggested transformation rules for the two labels are given in Figure 5.5. The transformation rule with kara in Figure 5.5 says that

<i>TAM label</i>	<i>Transformation</i>
kara	Karta must not be present. Karma is optional.
naa	Karta and karma are optional.
taa_huaa	Karta and karma are optional.

Figure 5.5: More transformation rules (for complex sentences)

karta of the verb with TAM label kara must not be present in the sentence and the karma is optionally present.

By these rules, the intermediate verb khaa (eat) in G.1 and kaaTa (cut) in G.2 do not have (independent) karta karaka present in the sentence. Ram is the karta of the main verb. phala (fruit) is the karma of khaa in G.1 but not of kaaTa in G.2. In the latter, phala is the karma of the main verb. All these are accommodated by the above transformation rule for ‘kara’.¹² The modifier-modified tree structures produced are shown in Figure 5.6 (ignore dotted lines).

5.6.2 Karaka Sharing

Now we give rules for obtaining the shared karakas.

Karta of the intermediate verb khaa in G.1 and G.2 can be obtained by the sharing rule S1’.

Rule S1’: Karta of a verb with TAM label ‘kara’ is the same as the karta of the main verb.

The sharing rule(s) are applied after the tentative karaka assignment is over using karaka to vibhakti mapping.

For karma of intermediate verb with TAM label kara, we have the sharing rule S2’.

Rule S2’: If an intermediate verb with the TAM label ‘kara’ takes a karma (as specified in its default karaka chart) while none has been obtained using karaka vibhakti mapping, then it shares its karma with the karta or the karma of the main verb.

¹²Bhartrahari formulated the original rule that if a nominal has a karaka relationship with two verbs, then the main verb will control its vibhakti.



Figure 5.6: Modifier-modified relations for sentences G.1, G.2 and G.3, respectively. (Shared karakas shown by dotted lines.)

With the above rules, the shared karakas for G.1, G.2 and G.3 are shown in Figure 5.6 by dotted lines. In the following example,

H.1 khariidakara raama ne phala khaayaa.
 having-bought Ram -ne fruit ate
 (Ram ate fruit having bought it.)

karta and karma of khariida (buy) are the same as the karta and karma, respectively, of khaa (eat). Figure 5.7 shows the relationships pictorially.

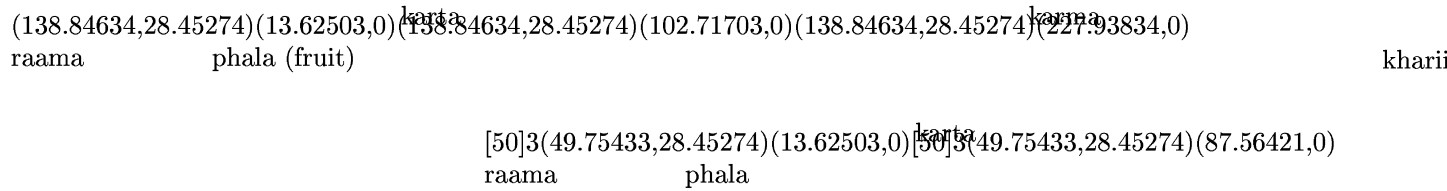


Figure 5.7: Parse structure for sentence H.1

Similarly, the sharing rule for taa_huaa can be stated as S3'.

Rule S3': An intermediate verb with TAM label 'taa_huaa' shares its karta with the karta of the main verb.

As example, consider:

H.2 shikaarii ne bhaagate hue shera ko dekhaa .
 hunter -ne while-running lion -ko saw
 (The hunter saw a lion while running.)

The above is ambiguous between whether the hunter was running or the lion was running. Now, karaka vibhakti mapping yields the following relation:

dekha (see)
 karta: shikaarii (hunter)
 karma: shera (lion).
 bhaaga (run)
 no karaka relation expressed explicitly

The above are shown pictorially in Figure 5.8 (ignore dotted lines for now).

The part of parse structure shown by solid lines is obtained by karaka vibhakti mapping. The part shown in dotted lines (the karta of bhaaga) is obtained by Rule S4.

When there are more than one intermediate verb groups in a sentence, there is a need to generalize the rules given earlier. To obtain the shared karaka of an intermediate verb, instead of getting it from the main verb we must obtain it from the verb (action) modified by the intermediate verb. The modified verb could be the main verb or another intermediate verb. The new generalized Rules are S1, S2 and S3.

Rule S1: Karta of intermediate verb with TAM label ‘kara’ is the same as the karta of the verb modified by the intermediate verb.

Rule S2: If an intermediate verb with the TAM label ‘kara’ takes a karma (as specified in its default karaka chart) while none has been obtained using karaka vibhakti mapping, then it shares its karma with the karta or the karma of the verb modified by the intermediate verb.

Rule S3: An intermediate verb with TAM label ‘taa_huaa’ shares its karta with the karta of the verb modified by the intermediate verb.

As an illustration, consider the following sentence from Hindi:

H.3 raama ne haatha se chiilkara kelaa khaate hue
Ram -ne hands -se having-peeled banana eating

bandara ko dekhaa.
monkey -ko saw.
(Ram saw a monkey eating a banana having peeled it
with his hands.)

The above is ambiguous as to who peeled and ate the banana. Karaka to vibhakti mapping produces the following relations:

```
dekha (see)
  karta: raama
  karma: bandara (monkey)
chiila (peel)
  karana: haatha (hand)
khaa (eat)
  karma: kelaa (banana)
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The above relations are shown in solid lines in Figure 5.10. Now, on applying the karaka sharing rule we get the following four alternative parses:

1. Assume chiila (peel) modifies khaa (eat).¹³ By Rule S1, karta of chiila is the same as the karta of khaa. Similarly, karma of chiila is the same

¹³‘chiila’ has the TAM label ‘kara’, which means that it precedes the action it modifies. Thus, ‘chiila’ modifies ‘khaa’ means that the action ‘chiila’ (peel) was performed before performing the action ‘khaa’ (eat).

as the karma of khaa, by Rule S2. Now, there are two alternatives for what khaa modifies.

- (a) khaa (eat) modifies the verb dekha (see). By Rule S3, karta of khaa is the same as the karta of dekha (see). Therefore:

```
khaa (eat)
  karta: raama
chiila (peel)
  karta: raama
  karma: kelaa (banana)
```

Figure 5.10 (a) shows the parse structure.

- (b) khaa (eat) modifies the noun bandara (monkey). By Rule S4, bandara is the karta of khaa (eat).

```
khaa (eat)
  karta: bandara (monkey)
chiila (peel)
  karta: bandara (monkey)
  karma: kelaa (banana)
```

Figure 5.10 (b) shows the parse structure.

2. Assume chiila (peel) modifies dekha (see). By Rule S1, karta of chiila is same as karta of dekha, namely raama. By Rule S2, the karma of chiila (peel) is the same as karma of dekha (see), namely, bandara (monkey). This is semantically anomalous as monkeys cannot be peeled. If world knowledge shows this to be an anomaly, further processing would be blocked. However, if such world knowledge is not available to the system, it would continue processing. There would be two alternatives. Karta of khaa can have two possible choices like in 1(a) and 1(b).

- (c) khaa (eat) modifies dekha (see). Therefore, karta of khaa is the same as the karta of dekha. (See Figure 5.10(c).)

```
khaa (eat)
  karta: raama
chiila (peel)
  karta: raama
  karma: bandara
```

The kartas happen to be the same as in 1(a).

- (d) khaa (eat) modifies bandara (monkey), which by rule S4 makes it the karta of khaa. (See Figure 5.10(d).)

```
khaa (eat)
  karta: bandara (monkey)
```

```

chiila (peel)
  karta: raama
  karma: bandara

```

5.7 Summary

In this chapter, we have discussed the linguistic aspect of Paninian framework as applied to modern Indian languages.

```

(238.05309,28.45274)(13.62503,0)karta
raama      bandara(monkey)
(238.05309,28.45274)(94.18903,0)(238.05309,28.45274)karma

```

```

(157.48909,28.45274)(13.62503,0)karta
raama      kelaa(banana)
(157.48909,28.45274)(86.53622,0)(157.48909,28.45274)karma

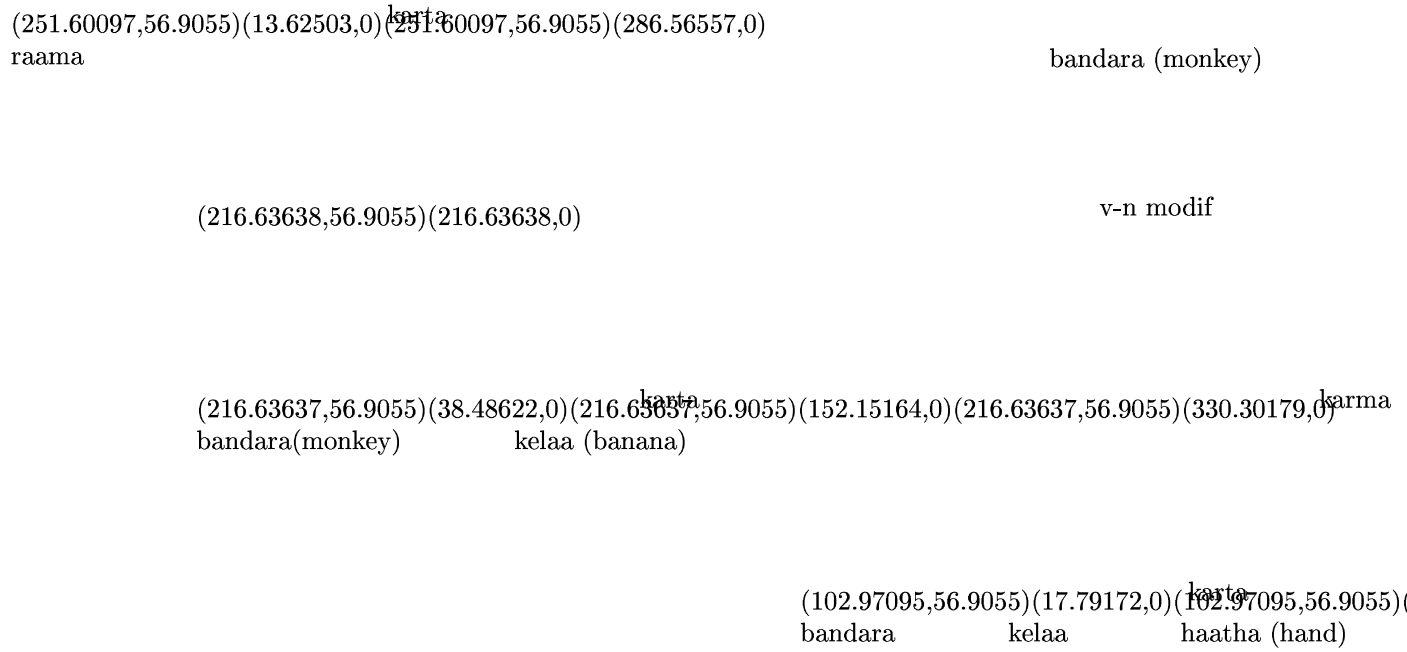
```

```

(84.57788,28.45274)(13.62503,0)karta
raama      kelaa      haatha (hand)
(84.57788,28.45274)karma

```

(a)



(b)

Figure 5.10: Alternative parses (a) and (b) for the sentence H.3

(248.06699,56.9055)(13.62503,0) ^{karta} (248.06699,56.9055)(73.49452,0) ^{karta} (180.09047,0) ^{karta} (248.06699,56.9055)(382.06699,56.9055)
 raama bandara khaa (eat)

(60.35147,56.9055)(13.62503,0) ^{karta} (60.35147,56.9055)(13.62503,0) ^{karta} (113.61963,56.9055)(95.85568,0)
 raama kelaa (banana) raama bandara (monkey) haatha (hand)

(c)

(240.07675,56.9055)(11.12503,0) ^{karta} (240.07675,56.9055)(137.58214,0) ^{karta} (366.53386,0) ^{karma}
 rama bandara(monkey)

(86.87933,56.9055)(86.87933,0) ^{v-n modif} (113.61963,56.9055)(13.62503,0) ^{karta} (113.61963,56.9055)(95.85568,0)
 raama bandara (monkey) khaa (eat) haatha (hand)

(86.87932,56.9055)(40.15288,0) ^{karta} (86.87932,56.9055)(141.25858,0) ^{karma}
 bandara (monkey) kelaa (banana)

(d)

Figure 5.10: Alternative parses (c) and (d) for the sentence H.3

The original framework was applied to Sanskrit; here the concepts and spirit of the framework have been used in explaining various language phenomena.

Further Reading

Ideas presented in this chapter have evolved over the years, earlier presented in Bhanumati (1989), Bharati et al. (1990), (1990b), (1991a), (1993a), etc.

See Appendix A for a discussion on how Paninian Grammar is applied to Sanskrit. References to traditional Paninian grammarians including Panini and Bhartrahari are also given there. Iyer() discusses Bhartrahari's work on means.

Exercises

5.1 In the Paninian grammar, why do you need karaka chart transformation? Illustrate with example sentences from, say, Hindi.

5.2 Show karaka assignments for the following sentences with and without shared karakas. Show the rules by which shared kaarakas are identified. (In case of ambiguity show more than one assignment)

- (a) raama khaanaa khaakara paanii piitaa huuaa ghara gayaa.
 Ram food having-eaten water drinking home went
 (Having eaten food, Ram went home drinking water.)
- (b) raama dauDataa huuaa murgaa khaakara ghara gayaa.
 Ram running chicken havind-eaten home went
 (Having eaten a chicken, Ram went home running. OR
 Having eaten a running chicken, Ram went home.)

5.3

(a) Show the modifier-modified tree with karaka relations, verb-verb relations, etc. for the sentence given below:

raama ne beiThakara caavala pakaane ke liya
 Ram -ne after-sitting rice to-cook

rasoiye ko patiilaa diyaa.
 cook -ko vessel gave
 (Ram gave a vessel to the cook for cooking rice
 after sitting down.)

Show all parses for the sentence. A parse which cannot be ruled out on the basis of syntax should be shown. (English translation is shown for convenience. It might, however, not show all parses.)